PAAWAN DESAI

paawankdesai@gmail.com ✤ linkedin.com/in/paawan-desai ✤ paawankdesai.com ✤ github.com/paawandesai

EDUCATION

The University of Texas at Austin - Cockrell School of Engineering **Graduated: May 2025**
Bachelors in Science in Electrical and Computer Engineering ✤ Minor in Entrepreneurship
College Coursework: Embedded Systems, Algorithms, Software Engineering, Data Science Lab, Applied Linear Algebra, Computer Vision, Data Analytics, Edge AI
Skills: Python, C, C++, Java, SQL, HTML/CSS, JavaScript (React.js), MongoDB, TensorFlow, PyTorch, scikit-learn, XGBoost, MATLAB, LangGraph, LangChain, OpenCV, Verilog, ARM, Edge AI, Time-Series Forecasting, Signal Processing, Energy Systems Integration, Microgrids

EXPERIENCE

AI Training & Evaluation **June 2026 - Present**

Handshake AI - Electrical and Software Engineer (Contract) **June 2026 - Present**
- ➔ Developed and evaluated complex, domain-specific engineering prompts to train large language models (LLMs) on electronic design automation (EDA).
- ➔ Engineered virtual simulation environments and workflows for circuit design software (LTspice, ngspice, KiCad, Qucs) to generate high-quality AI training data.
- ➔ Analyzed LLM outputs for technical accuracy, algorithmic efficiency, and advanced engineering communication capabilities.
- ➔ Contributed to improving AI reasoning in hardware simulation and software execution through expert review and targeted feedback loops.

Micro1 - Software Engineer (Contract) **May 2026 - Present**
- ➔ Contributed software engineering domain expertise to train and evaluate code-generation capabilities of foundational large language models.
- ➔ Executed testing of AI-generated code by designing prompts, running execution environments, and validating algorithmic correctness.
- ➔ Reviewed model outputs for technical accuracy, edge-case handling, and software best practices to improve overall reasoning quality in AI training pipelines.

Mercor - Software Engineer (Contract) **Nov 2025 - Present**
- ➔ Engineered and deployed automated tooling infrastructure to train, evaluate, and validate complex API behaviors for enterprise AI models.
- ➔ Designed testing frameworks to programmatically verify AI-generated system responses, improving pipeline reliability and output accuracy.
- ➔ Conducted quality assurance and data evaluation for multi-modal AI generation pipelines

SkyTerra Group - AI & Data Analyst **April 2025 - Nov 2025**
- ➔ Engineered biometric AI pipelines for a healthcare client, training ML models to detect ECG-based cardiac abnormalities from rhythm data
- ➔ Designed HIPAA- and GDPR-compliant edge-to-cloud data architectures, enabling secure, scalable real-time health monitoring
- ➔ Led field integration of a hybrid microgrid, deploying battery systems, inverters, generators, and remote monitoring for an off-grid client site

Fundare Finance - CoFounder and Engineer **May 2024 - Dec 2024**
- ➔ Lead the development of a Python and SQL-powered dashboard to streamline revenue growth tracking for small businesses

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PROJECTS/RESEARCH

AI Agent Red Team Engine - Adversarial Security Research **April 2026**
- ➔ Built an adversarial penetration testing engine that executes multi-turn social engineering and RAG injection attacks against AI agents
- ➔ Mapped 113 adversarial prompts across 8 attack categories to OWASP LLM Top 10, MITRE ATLAS, CWE, and EU AI Act compliance
- ➔ **Tools:** Python, LangGraph, Anthropic API, OpenAI API, Pydantic, FastAPI, asyncio, ChromaDB
- ➔ **Open Source:** https://github.com/paawandesai/redteam-engine

Social Impact BuildFest – EveryLibrary Immersive Advocacy Project **Feb 2025**
- ➔ Built a Python GUI integrating sentiment analysis (NLTK, Hugging Face) and text-to-speech to drive haptic feedback for DataFeel devices.
- ➔ Enhanced accessibility with dyslexia-friendly fonts and UI customizations
- ➔ **Tools:** Python, PyQt6, NLTK, Hugging Face, Pyttsx3, Matplotlib, DataFeel SDK

Energy A.I. Hackathon 2025 – First Place **Jan 2025**
- ➔ Built a predictive pipeline to forecast Grid, Diesel, and CNG energy usage for hydraulic fracturing, incorporating uncertainty modeling.
- ➔ Implemented XGBoost and Random Forest models with R^2: 0.9918 (Grid), 0.9766 (Diesel), 0.9692 (CNG)
- ➔ **Tools:** Python, Jupyter Notebook, scikit-learn, XGBoost, Pandas, NumPy, CatBoost, ozbayrak_goodness, UTuning

Hardware Accelerator for Post-Quantum Cryptography (ML-DSA) - Senior Project **Sep 2024 - May 2025**
- ➔ Led development of Verilog accelerator to enhance cryptographic operations for the Module-Lattice-Based Digital Signature Algorithm
- ➔ Designing key modules like a random bit generator and NTT while creating a C-based driver for hardware-software integration
- ➔ **Tools:** Verilog, ModelSim, C, Python

Urban Insights Initiative **Jan 2024 - May 2024**
- ➔ Automated site analysis for urban planning using OSMnx and OpenCV, creating detailed maps of building footprints, greenery, and traffic
- ➔ **Tools:** Python, OpenCV, OSMnx, NetworkX

Human Centered Robotics Laboratory - Adaptable/Deployable Structures **Summer 2022**
- ➔ Engineered a deployable prototype transforming a truss-frame system into a vertical-link eccentrically braced frame for seismic resistance
- ➔ Simulated structural behavior under seismic loads using MATLAB and validated mechanisms with LabVIEW
- ➔ **Tools:** MATLAB, LabVIEW, SolidWorks

Honors and Leadership

Honors:
Achieved Top 2.5% in American Mathematics Contest (AMC); AIME Qualifier (9/15), Secured 3rd Place Nationally in the American Regional Mathematics League, Awarded 2nd Place in the Texas Alliance for Minorities in Engineering competition
IEEE Robotathon

- ➔ Designed and built a robot using Arduino, earning 2nd place in the Mechanical Design competition

TEDxYouth@Austin (Volunteer Role)

- ➔ Co-organized the largest global TEDxYouth event, securing $30k in sponsorships and hosting 1,100+ attendees

Captain of Punjabbawockeez

- ➔ Led performances, finances, and community service initiatives, raising funds for cancer research through Texas Thon